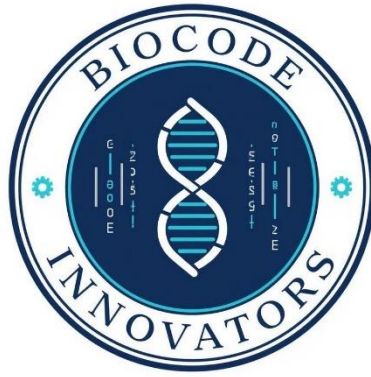




BioCode Innovators

Bioinformatics Bootcamp

ASSESSMENT: 02

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Question 1

A molecular biologist has been provided with only a gene accession number from a published research article and needs to retrieve the corresponding DNA sequence, protein sequence, associated literature, and gene information.

Describe the step-by-step approach you would follow using NCBI resources. Explain which databases and tools you would access during this process and why.

Answer:

When you start a bioinformatics search with only a gene accession number, NCBI is the most suitable platform because it relies on a unified cross reference system called Entrez. The accession number itself is a unique identifier, meaning it acts like a secure tracking code that eliminates ambiguity and leads directly to one specific biological entry. By pasting this ID into the global Entrez search bar, the system simultaneously queries separate databases, including those for genes, nucleotides, proteins, and scientific literature, saving the researcher from having to search each database manually.

Once the results are generated, the researcher can click into the NCBI Gene or Nucleotide databases to extract the raw molecular blueprint of the target. For sequence analysis, downloading the data in a FASTA format is ideal because its simple plain text structure is perfectly optimized for downstream computer tools like BLAST sequence alignments. If the researcher needs deeper historical context, regulatory information, or specific exon-intron boundaries, they can switch to the GenBank format, which contains rich text annotations explaining the specific anatomy of the sequence.

Finally, the integrated nature of NCBI allows the user to seamlessly jump to protein data and published articles without starting a new search. Because genes directly code for proteins, the nucleotide page contains embedded cross references that link straight to the translated amino acid sequence in the NCBI Protein database. Additionally, every verified sequence entry is tied to a bibliography section linked to PubMed. This allows the researcher to instantly retrieve the original peer-reviewed literature associated with the gene, turning a simple tracking number into a complete biological profile.

Question 2

A structural biologist is interested in studying the three-dimensional structure of the human insulin protein to understand how mutations may affect its function. Which database would you recommend for this purpose? Explain how you would retrieve the structure and describe the information available within the database that would support the research.

Answer:

For a structural biologist studying how mutations physically alter human insulin, the Protein Data Bank (PDB) is the definitive database choice. While genomic repositories like NCBI excel at archiving raw genetic code, they cannot display how a protein folds or interacts in physical space. The PDB is a dedicated structural database that houses experimentally validated three dimensional atomic coordinates. This is scientifically essential because a protein's cellular function is entirely dictated by its physical shape, and seeing the molecule in 3D is the only way to track structural changes caused by genetic variations.

To retrieve this data, the researcher inputs "human insulin" into the PDB search engine. When selecting a specific entry, it is critical to evaluate the experimental resolution, aiming for a low value such as 1.5 Angstroms. In structural biology, a lower resolution number signifies a sharper, higher quality electron density map. This ensures that the spatial positioning of individual atoms is highly reliable, which prevents false positive conclusions and gives the researcher total confidence in the accuracy of the molecular model.

Within the entry page, the PDB provides interactive visualization tools and precise data regarding ligand binding sites. This information directly supports mutation research by allowing a scientist to observe if an amino acid swap occurs at a critical spot, such as the disulfide bonds that pin the insulin chains together. By comparing normal insulin with mutated variants, researchers can visually track how minor structural distortions prevent insulin from fitting into cell receptors, mapping physical shape changes directly to clinical conditions like diabetes.

Question 3

Imagine you have been assigned a protein with no prior knowledge except its accession ID. Design a complete workflow describing how you would retrieve, analyze, and interpret all

available information about this protein using NCBI, UniProt, and the Protein Data Bank (PDB). Explain the purpose of each step and how the information from one database complements the others.

Answer:

Analyzing a completely mystery protein using only an accession number requires a structured workflow that builds sequentially from raw sequence data to functional and structural understanding. The process begins at NCBI, which operates as a massive primary database. By entering the accession number into the Entrez search engine, you retrieve the foundational genetic or amino acid sequence. Downloading this sequence in a FASTA format is essential because its simple text structure is the universal standard needed to run comparative alignment algorithms and figure out the evolutionary origin of the molecule.

Once the sequence is secured, the next phase shifts to UniProt, which serves as a premier secondary database. While NCBI tells us exactly what the sequence is, UniProt explains what the sequence actually does. It achieves this by providing curated, interpreted functional annotations and standardized Gene Ontology vocabularies. By analyzing this layer of data, you can quickly uncover the protein's metabolic pathways, discover its specific cellular location, and understand its physiological roles without having to sort through unorganized raw data.

The final phase brings the workflow full circle by tracking cross-references from UniProt over to the Protein Data Bank (PDB). This structural step is critical because knowing a protein's theoretical function is incomplete without visualizing its physical machinery. The PDB entry allows you to view the 3D model of the protein, matching the active binding sites and physical pockets to the cellular jobs discovered in UniProt. Ultimately, these three databases complement each other perfectly, transforming a single string of anonymous text into a fully realized, three-dimensional model of biological reality.

PART II

Choose one biological sequence (DNA or Protein) from NCBI, UniProt, or the Protein Data Bank (PDB). Include the following in your submission:

- Screenshot of the retrieved database record.
- Accession ID or Entry ID of the selected sequence.
- Name of the Gene/Protein.
- Source Organism.
- A brief description (150–250 words) explaining:
 - o What the sequence or protein is.
 - o Its biological function.
 - o Why it is important.
 - o Which database you used and why you selected it.

Answer:

Selected Sequence:

- Accession ID: P42212
- Name of the Protein/Gene: Green Fluorescent Protein/*GFP* gene
- Source Organism: Water jellyfish

Screenshot:

The screenshot displays the UniProt database record for the protein P42212, identified as GFP_AEQVI. The interface includes a top navigation bar with links to BLAST, Align, Peptide search, ID mapping, SPARQL, and UniProtKB. The main content area is divided into a left sidebar with navigation links (Function, Names & Taxonomy, Subcellular Location, Phenotypes & Variants, PTM/Processing, Expression, Interaction, Structure, Family & Domains, Sequence, Similar Proteins, Homologs) and a central panel. The central panel shows the protein's name, accession number, and a table of key information: Protein (Green fluorescent protein), Gene (GFP), Status (UniProtKB reviewed (Swiss-Prot)), Organism (Aequorea victoria (Water jellyfish) (Mesonema victoria)), Amino acids (238), Protein existence (Evidence at protein level), and Annotation score (5/5). Below this, there are tabs for Entry, Variant viewer, Feature viewer, Genomic coordinates, Publications, External links, and History. The 'Function' tab is active, showing a description of the protein's role as an energy-transfer acceptor and its biotechnological applications. The 'Biotechnology' section describes how GFP has been engineered into various mutants and fusion proteins for research and industrial use.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search Help

P42212 · GFP_AEQVI

Protein ⁱ	Green fluorescent protein	Amino acids	238 (go to sequence)
Gene ⁱ	GFP	Protein existence ⁱ	Evidence at protein level
Status ⁱ	UniProtKB reviewed (Swiss-Prot)	Annotation score ⁱ	5/5
Organism ⁱ	Aequorea victoria (Water jellyfish) (Mesonema victoria)		

Tools Download Add Add a publication Entry feedback

Entry Variant viewer Feature viewer Genomic coordinates Publications External links History

Functionⁱ

Energy-transfer acceptor. Its role is to transduce the blue chemiluminescence of the protein aequorin into green fluorescent light by energy transfer. Fluoresces in vivo upon receiving energy from the Ca²⁺-activated photoprotein aequorin.

Biotechnologyⁱ

Green fluorescent protein has been engineered to produce a vast number of variously colored mutants, fusion proteins, and biosensors. Green fluorescent protein can be mutated to emit at different wavelengths such as blue for BFP (when Tyr-66 is replaced by His), cyan for CFP (when Tyr-66 is replaced by Trp), and yellow for YFP (when Thr-203 is replaced by Tyr). Further generation of mutants led to more stable proteins (at 37 degrees Celsius for example) with brighter fluorescence and longer fluorescence lifetimes. Fluorescent proteins and their mutated allelic forms have become a useful and ubiquitous tool for making chimeric proteins, where they function as a fluorescent protein tag. Typically they tolerate N- and C-terminal fusion to a broad variety of proteins. They have been expressed in most

Green Fluorescent Protein (GFP) is a specialized luminescent protein natively found in the translucent tissue of the jellyfish *Aequorea victoria*. Scientifically, its natural function is to act as an energy transfer acceptor. It absorbs blue chemiluminescence emitted by a companion jellyfish protein called aequorin and transforms it into a brilliant green glowing light. What makes GFP unique is its completely self contained design; its internal light emitting core, or chromophore, forms automatically out of its own amino acid chain without needing any complex cellular machinery or helper enzymes to function.

In modern biomedical research, GFP serves as a revolutionary molecular tool that allows scientists to visually track invisible microscopic activity inside living cells. By fusing the genetic code of GFP to a target human protein, researchers can watch live cellular transport, monitor gene expression changes, and trace virus distributions under a microscope in real time.

UniProtKB/Swiss-Prot was chosen as the most suitable database for retrieving this sequence because it is a highly reliable secondary repository. While primary databases like NCBI hold vast numbers of raw, unedited sequence letters, UniProt provides manually curated biological context. This expert screening ensures that all functional annotations, active site mutations, and disease linkages are fully verified. This effectively eliminates false-positive results, giving researchers complete confidence in the accuracy of their downstream analysis.